# Topic Selection and Emotional Tone in Ideologically Different Korean Newspapers on Twitter [1]

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Digital Korea

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[1] Replication package—including the data, analysis workflow, exported figures, and data dictionary—is publicly available at: https://github.com/nikster4/NvDriel-DigitalKorea-Final-paper

## Introduction

Social media platforms have become an increasingly important channel through which newspapers distribute news and engage with audiences.[2] Unlike traditional newspaper articles, social media posts are constrained by platform design and audience expectations. Twitter in particular encourages short-form communication, rapid dissemination of information, and competition for user attention.[3] As a result, newspapers may adapt their communication style to fit the platform environment.

This raises questions about how ideological differences among newspapers manifest on social media. While previous research has demonstrated ideological variation in news coverage and political communication, it is less clear whether such differences remain visible when newspapers communicate through short-form social media content.[4] Platform incentives may encourage newspapers to adopt similar communication styles even when their political orientations differ. At the same time, ideological differences may continue to influence which issues newspapers prioritize and how those issues are presented to audiences.

South Korea provides a particularly useful case for examining this question because its newspaper landscape contains well-established ideological distinctions between progressive and conservative outlets.[5] While these differences have been documented in studies of traditional news coverage, less is known about how they manifest in social media communication. Because tweets function more like headlines than full articles, they provide an opportunity to examine whether platform norms encourage convergence across ideological groups.

This paper examines tweets from six South Korean newspapers classified as either left-leaning or right-leaning. The study applies topic modelling and dictionary-based sentiment analysis to investigate differences in both topic selection and emotional tone.

The research question guiding this study is:

**How do ideological differences among Korean newspapers on Twitter manifest in topic selection and emotional tone?**

Based on existing research on ideological media differences, I expect topic selection to differ more strongly than emotional tone between ideological groups.

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[2] Bastos, M. T. (2015). Shares, Pins, and Tweets: News readership from daily papers to social media. *Journalism Studies (London, England)*, *16*(3), 305–325. https://doi.org/10.1080/1461670X.2014.891857

[3] Carlson, M. (2018). Facebook in the News: Social media, journalism, and public responsibility following the 2016 Trending Topics controversy. *Digital Journalism*, *6*(1), 4–20.

[4] Bastos,Shares, Pins, and Tweets, 305–325.

[5] Kim, Byung Wook. (2020). *The political economy of media framing in Korea : An analysis of Korean news coverage of climate change, 1995-2015* [University of Iowa].

## Literature Review

Computational text analysis has become an increasingly important approach within the social sciences, allowing researchers to analyze large collections of text that would be difficult to examine manually. Rather than replacing close reading, computational methods enable researchers to identify broader patterns, recurring themes, and relationships within large corpora. Topic modelling is particularly useful because it can uncover latent thematic structures within a collection of documents, allowing researchers to compare how different groups discuss similar issues.[6]

One of the most widely used topic modelling techniques is Latent Dirichlet Allocation (LDA). LDA assumes that documents contain mixtures of topics and that topics themselves consist of recurring groups of words. By identifying which words frequently occur together, LDA can reveal underlying themes within a corpus that may not be immediately apparent through manual reading alone.[7] Because topic models focus on patterns across large numbers of documents, they are especially useful for studies comparing different media outlets or ideological groups.

At the same time, researchers have emphasized that computational text analysis is heavily influenced by preprocessing decisions. Denny and Spirling argue that seemingly minor choices regarding tokenization, stopword removal, stemming, and text cleaning can substantially affect the results produced by unsupervised learning methods. As a result, preprocessing should not be viewed as a purely technical step but as an important methodological decision that must be documented transparently.[8] Their work highlights the importance of carefully describing data preparation procedures when conducting topic modelling and other forms of computational text analysis.

The rise of social media has transformed the way news is produced, distributed, and consumed. Traditional newspapers no longer operate solely within print or even website-based environments; they increasingly rely on social media platforms to reach audiences. As social media platforms have become central gateways to news consumption, newspapers have adapted their communication strategies to fit platform-specific expectations and constraints. Research has shown that social media increasingly shape patterns of news readership and influence which stories gain visibility among audiences.[9]

Importantly, social media platforms differ from traditional news environments because audience engagement plays a much larger role in determining visibility. Bastos found that social media users do not necessarily prioritize the same topics as newspaper editors, demonstrating that the distribution of attention on social media can differ substantially from editorial priorities. While newspapers emphasize certain topics through editorial decisions,

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[6] Grimmer, Justin, Margaret E. Roberts, and Brandon M. Stewart. Text as data: A new framework for machine learning and the social sciences. Princeton University Press, 2022.

[7] Grimmer, Justin, Roberts, and Stewart, Chapter 13: Topic Models.

[8] Denny, M. J., & Spirling, A. *Text preprocessing for unsupervised learning: Why it matters, when it misleads, and what to do about it* (Political Analysis, 26(2), 2018), 168-189.

[9] Bastos, Shares, Pins, and Tweets, 305–325.

social media audiences selectively share and engage with content in ways that create different patterns of visibility.[10] This suggests that newspapers operating on social media may face incentives to adapt their communication styles in order to maximize engagement.

The growing importance of social media has also generated concerns regarding the role of algorithms and platform design in shaping public discourse. Carlson argues that social media platforms increasingly function as important actors within the news ecosystem, influencing how news is distributed and consumed. Rather than simply serving as neutral channels, platforms actively shape visibility through engagement metrics, recommendation systems, and algorithmic curation.[11] This has led researchers to question whether platform logics encourage particular styles of communication, including emotionally engaging, simplified, or attention-grabbing forms of news presentation.

These developments are particularly relevant for journalism on Twitter. The platform's emphasis on brevity, rapid circulation, and engagement metrics encourages communication styles that differ from those traditionally associated with newspaper reporting. Because tweets function more like headlines than full articles, newspapers must communicate information within severe space constraints while simultaneously competing for user attention. This creates the possibility that newspapers with different ideological orientations may converge toward similar communication styles as they adapt to the demands of the platform.

At the same time, ideological differences remain an important feature of news production. Research on Korean media has consistently identified ideological distinctions between conservative and progressive newspapers. Although these outlets operate within the same media environment, they often differ in issue selection, framing, and editorial priorities. Kim's analysis of Korean newspaper coverage demonstrates that meaningful differences can be observed between conservative and progressive newspapers even when they cover similar issues.[12] This suggests that ideological distinctions may continue to influence content selection even when newspapers communicate through social media platforms.

Taken together, this literature suggests two competing possibilities. On the one hand, ideological differences may continue to shape how newspapers communicate, resulting in distinct patterns of topic selection and emotional tone. On the other hand, the communication norms of Twitter may encourage convergence, pushing newspapers toward similar styles regardless of ideological orientation. By combining topic modelling and sentiment analysis, this study investigates how these competing influences manifest in a corpus of Korean newspaper tweets.

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[10] Bastos, Shares, Pins, and Tweets, 305–325.
[11] Carlson, Facebook in the News.
[12] Kim, *The political economy of media framing in Korea.*

## Data and Methods

### Dataset

The dataset consists of 2,745 tweets, posted in 2017, collected from the official Twitter accounts of six South Korean newspapers. The corpus contains 1,496 tweets from left-leaning newspapers and 1,249 tweets from right-leaning newspapers.

Because tweets are short texts, they resemble headlines more than full news articles. This makes the dataset particularly suitable for examining how newspapers adapt their communication strategies to social media environments.

### Preprocessing

Two preprocessing pipelines provided by the course instructor were used. Both scripts utilized KiwiPiePy for Korean language processing.

For the sentiment analysis, the tweets were processed using the instructor-provided sentiment analysis preprocessing script. Sentiment scores were then generated using the positive and negative KNU Word lists provided by the instructor. The sentiment analysis was performed using Orange3-3.40.0.

For topic modelling, a separate instructor-provided preprocessing script was used. The document frequency was turned off in comparison to the original provided script. Following preprocessing, stopword removal was applied before conducting topic modelling. Topic modelling was performed using Orange 3.39.0.

### Sentiment Analysis

Dictionary-based sentiment analysis was used to examine emotional tone. Sentiment distributions were first compared between left-leaning and right-leaning newspapers.

To investigate the relationship between engagement and sentiment, tweets were sorted according to retweet count. The top 25% and bottom 25% of tweets were selected, resulting in approximately 686 tweets in each subset. Sentiment analysis was then performed separately on each group.

### Topic modelling

Latent Dirichlet Allocation (LDA) was used to identify thematic patterns within the corpus.[13] The dataset was divided into left-leaning and right-leaning newspapers prior to topic modelling.

Several topic configurations were explored. A six-topic solution was ultimately selected because it produced more interpretable and distinct topics than alternative configurations while avoiding excessive overlap between topics.

[13] Grimmer, Roberts, and Stewart (Chapter 13).

## Analysis and Findings

### Emotional Tone Across Ideological Groups

The sentiment analysis revealed substantial similarity between left-leaning and right-leaning newspapers. Both groups displayed highly comparable sentiment distributions, with most tweets clustered around neutral sentiment values.

Although individual newspapers occasionally exhibited variation, the overall ideological comparison showed little evidence of systematic differences in emotional tone. This suggests that ideological orientation is not strongly reflected through sentiment within the Twitter corpus.

Figure 1 displays the sentiment distributions for left-leaning and right-leaning newspapers.

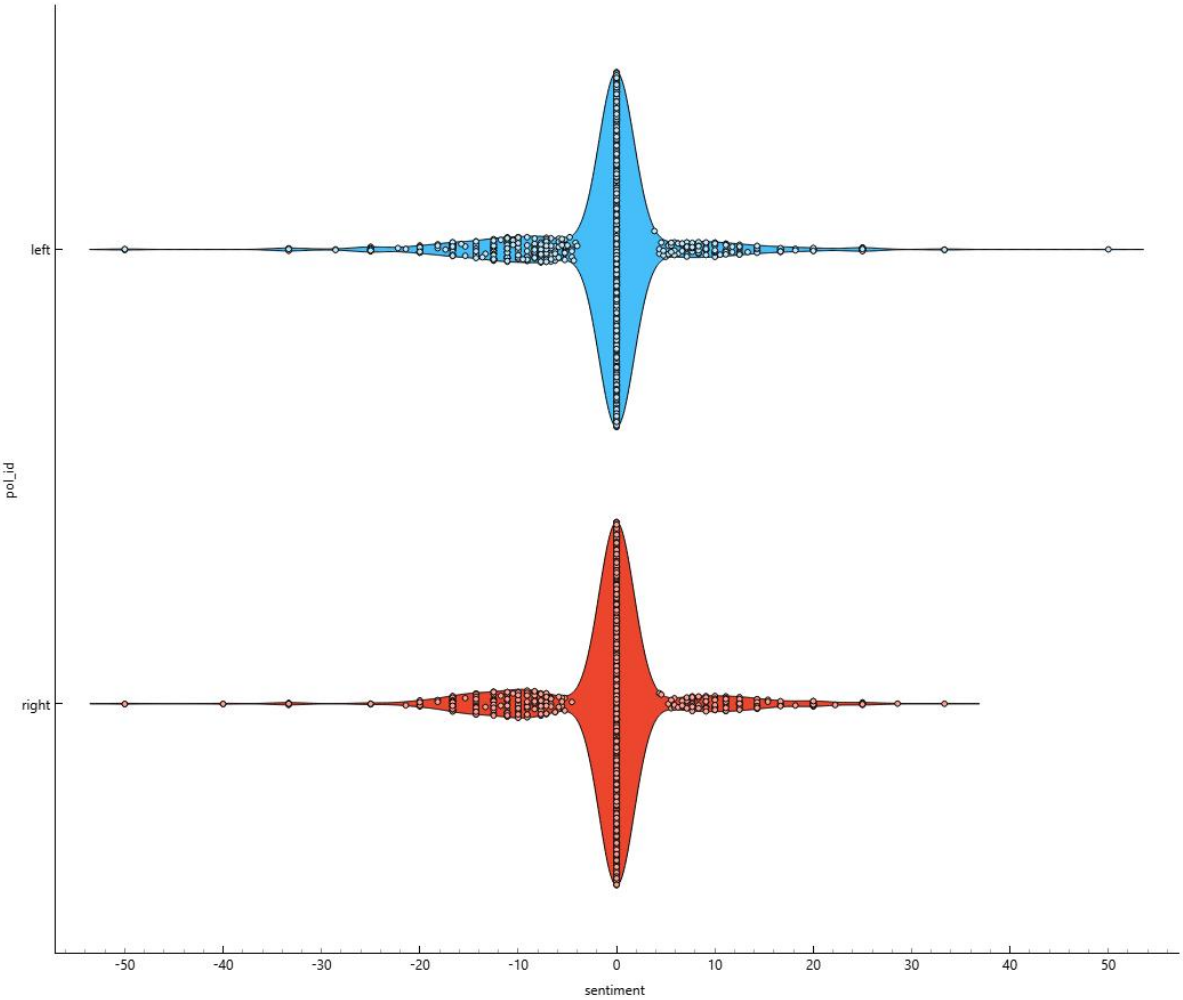


***Figure 1. Sentiment distribution of tweets from left-leaning and right-leaning newspapers. The distributions are highly similar, suggesting that emotional tone does not strongly differ between ideological groups.***

The similarity of the distributions indicates that newspapers from different ideological camps communicate with broadly comparable emotional intensity on Twitter.

### Topic Selection Across Ideological Groups

While sentiment appeared similar, topic modelling revealed more substantial differences in topic selection.

The six topics identified within the left-leaning corpus included themes related to political investigations, corruption scandals, labor and reform, North Korea, public policy, and the Moon administration. Topics frequently contained terms such as 검찰 (prosecution), 노동자 (worker), 개혁 (reform), 투기 (speculation), and 문재인 (Moon Jae-in).

The six topics identified within the right-leaning corpus emphasized foreign affairs, security issues, economic policy, minimum wage debates, government affairs, and relations with North Korea and China. Common terms included 사드 (THAAD), 북한 (North Korea), 최저임금 (minimum wage), 미사일 (missile), 중국 (China), and 정부 (government).

Although some overlap existed between the two ideological groups, particularly regarding government affairs and North Korea, differences emerged in issue emphasis. Left-leaning newspapers devoted greater attention to labor, reform, investigations, and political scandals. Right-leaning newspapers devoted greater attention to security issues, foreign affairs, and economic policy debates.

Figure 2 presents the topic modelling results.

| Topic | Topic keywords |
| --- | --- |
| 1 | 대통령, 정부, 그림판, 한겨레, 검찰, 북한, 대장, 국정원, 박찬주, 중국 |
| 2 | 대통령, 사람, 투기, 안철수, 박근혜, 공개, 여성, 피해자, 서울, 법원 |
| 3 | 대통령, 검찰, 트럼프, 청와대, 국민, 북한, 대표, 의원, 수사, 제재 |
| 4 | 정부, 문재인, 대통령, 사드, 평가, 배치, 대표, 환경, 여론, 교사 |
| 5 | 경향신문, 화백, 협의, 미국, 의원, 조사, 공관, 이재용, 박순, 장도리 |
| 6 | 대통령, 뉴스, 기업, 청와대, 지시, 노동자, 정부, 개혁, 국정, 사람 |

| Topic | Topic keywords |
| --- | --- |
| 1 | 한국, 미국, 동아일보, 뉴스, 가능, 정부, 국회, 대학, 북한, 대표 |
| 2 | 사드, 배치, 북한, 대통령, 교사, 뱅크, 카카오, 속보, 평가, 조작 |
| 3 | 대통령, 최저, 이유, 임금, 인상, 해외, 트럼프, 최고, 주장, 국내 |
| 4 | 대통령, 정부, 의원, 기업, 검찰, 사건, 공개, 엄마, 경찰, 미국 |
| 5 | 정부, 이유, 논란, 의원, 여성, 미국, 자유한국당, 대통령, 한국, 속보 |
| 6 | 대통령, 중국, 사람, 서울, 발사, 미사일, 이유, 장관, 의원, 올림픽 |

***Figure 2. Six-topic LDA results for left-leaning (top) and right-leaning (bottom) newspaper tweets. The topics reveal substantial overlap in broad issue areas such as government affairs and North Korea, while also indicating differences in issue emphasis, including labor and reform topics in left-leaning newspapers and economic and security-related topics in right-leaning newspapers.***

These findings suggest that ideological differences remain visible through the selection and prioritization of issues even when newspapers communicate through short-form social media content.

### Engagement and Emotional Tone

To examine whether emotional tone influences audience engagement, sentiment distributions were compared between the highest and lowest quartiles of retweeted tweets.

The results indicate a modest relationship between sentiment and engagement. Highly retweeted tweets exhibited somewhat greater negative sentiment than low-engagement tweets. However, the difference was relatively small and should not be overstated.

Figure 3 and 4 compare sentiment distributions between the top 25% (figure 3) and bottom 25% (figure 4) of tweets ranked by retweet count.

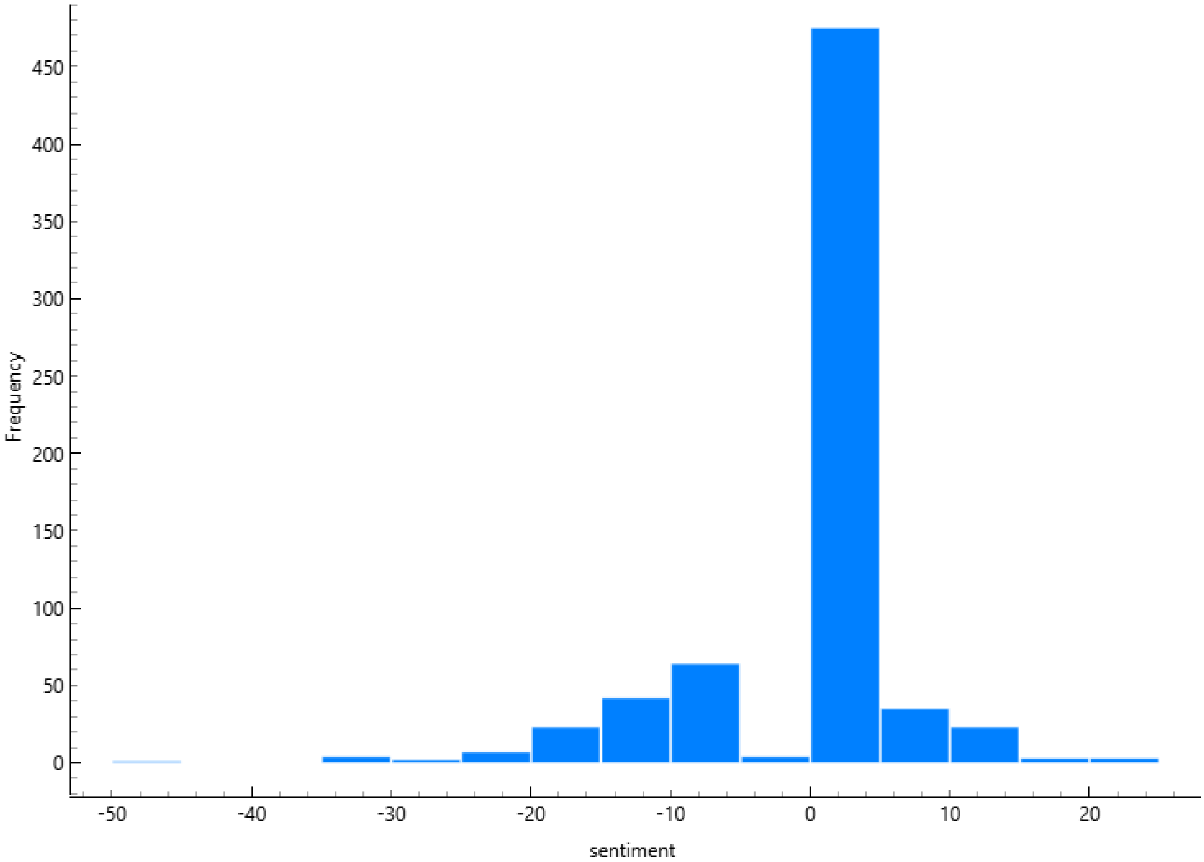


***Figure 3***

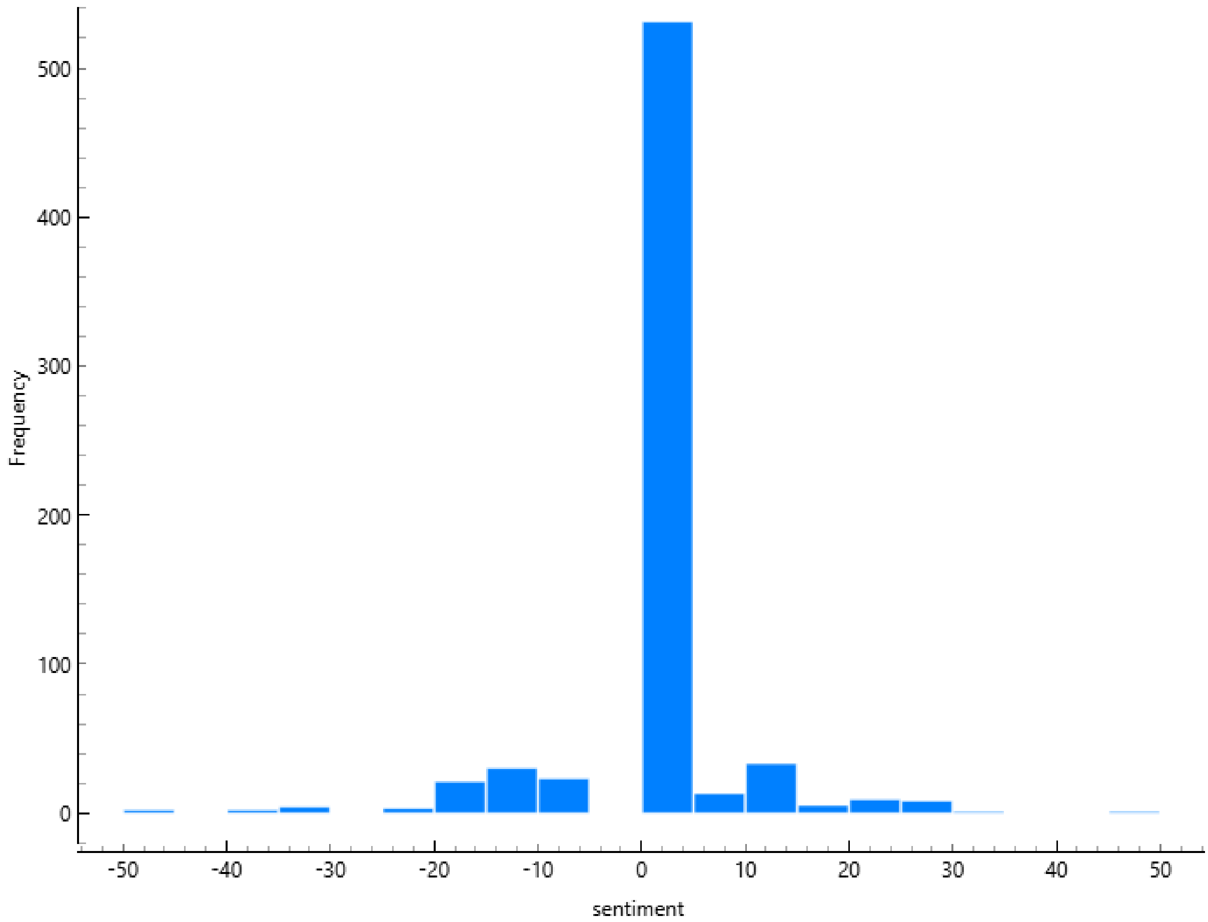


***Figure 4***

***Figure 3, 4. Sentiment distributions of the highest (top, figure 3) and lowest (bottom, figure 4) retweet quartiles. Highly retweeted tweets display a modest tendency toward more negative sentiment.***

The findings provide limited evidence that negativity may contribute to engagement. However, the effect appears substantially weaker than the differences observed in topic selection.

**Discussion**

The results suggest that ideological differences among Korean newspapers remain visible on Twitter, but primarily through topic selection rather than emotional tone.

The sentiment analysis revealed considerable convergence across ideological groups. Newspapers from different political orientations exhibited remarkably similar emotional profiles despite their differing editorial positions. This may reflect the influence of platform norms that encourage a broadly similar style of communication.

In contrast, topic modelling demonstrated that ideological distinctions remain visible through issue emphasis. Left-leaning newspapers were more likely to focus on labor, reform, and investigations, whereas right-leaning newspapers devoted greater attention to security issues, foreign affairs, and economic policy.

These findings suggest that newspapers may adapt to platform-specific communication styles while continuing to express ideological differences through content selection.

## Conclusion

This paper examined ideological differences among Korean newspapers on Twitter using topic modelling and dictionary-based sentiment analysis.

The findings indicate that ideological differences manifest more clearly through topic selection than through emotional tone. While left-leaning and right-leaning newspapers discussed different issues and emphasized different political concerns, their sentiment distributions were broadly similar.

The engagement analysis further suggested only a modest relationship between negative sentiment and retweet frequency. Although highly engaged tweets tended to exhibit slightly greater negativity, the effect was limited.

Overall, the results suggest that Twitter does not eliminate ideological differences among newspapers. Rather, ideological distinctions persist primarily through the issues that newspapers choose to discuss. At the same time, platform norms may encourage convergence in communication style, resulting in similar emotional tone across ideological groups.

Several limitations should be acknowledged. First, tweets are short texts and may not fully capture the editorial positions of newspapers. Second, dictionary-based sentiment analysis depends on the quality and coverage of the sentiment lexicon. Third, topic modelling requires subjective interpretation and alternative topic solutions may produce somewhat different results.

Future research could compare Twitter content with full newspaper articles to examine whether ideological differences become stronger or weaker across communication formats. Additional research could also investigate other forms of audience engagement, including likes and replies, to better understand how users respond to different forms of news communication.

# Bibliography

Bastos, M. T. (2015). Shares, Pins, and Tweets: News readership from daily papers to social media. *Journalism Studies (London, England)*, *16*(3), 305–325. https://doi.org/10.1080/1461670X.2014.891857

Carlson, M. (2018). Facebook in the News: Social media, journalism, and public responsibility following the 2016 Trending Topics controversy. *Digital Journalism*, *6*(1), 4–20. https://doi.org/10.1080/21670811.2017.1298044

Denny, M. J., & Spirling, A. *Text preprocessing for unsupervised learning: Why it matters, when it misleads, and what to do about it*. Political Analysis, 26(2), 168-189, 2018.

Grimmer, Justin, Margaret E. Roberts, and Brandon M. Stewart. Text as data: A new framework for machine learning and the social sciences. Princeton University Press, 2022.

Kim, Byung Wook. (2020). *The political economy of media framing in Korea : An analysis of Korean news coverage of climate change, 1995-2015* [University of Iowa]. https://doi.org/10.17077/etd.005219